

H185 THEOREM REVIEW

1.2.4 Thm. $f = u + iv$ on Ω open. f holo on $\Omega \Leftrightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
 Def. F primitive for f on Ω if F holo on Ω and $F' = f$ on Ω .

1.3.2 Thm. f has prim F in Ω , conn $\gamma: w_1 \rightarrow w_2 \Rightarrow \int_{\gamma} f(z) dz = F(w_2) - F(w_1)$
 Pf. $\int_{\gamma} f = \int_a^b f(z(t)) z'(t) dt = \int_a^b F'(z(t)) z'(t) dt = \int_a^b \frac{d}{dt} F(z(t)) dt = F(z(b)) - F(z(a))$

1.3.3 Cor. if γ closed, f has prim, $\int_{\gamma} f = 0$.

1.3.4 Cor. f holo in Ω , $f' = 0 \Rightarrow f$ constant.

2.1.1 Thm. (Cauchy's). $\Omega \subset \mathbb{C}$. $T \subset \Omega$. f holo $\Rightarrow \int_T f = 0$.

Pf. (i) keep splitting T so $|\int_T f| \leq 4^n |\int_{T_n} f|$.
 (ii) f holo at $z_0 \in T^{(n)} \Rightarrow f(z) = f(z_0) + f'(z_0)(z-z_0) + o(z-z_0)$
 (iii) $\Rightarrow |\int_{T_n} f| = |\int_{T_n} o(z)(z-z_0) dz| \leq \epsilon_n \text{diam}(T_n) \text{perim}(T_n)$
 where $\epsilon_n = \sup_{T_n} |o(z)|$. $n \rightarrow \infty$ shows $\int_T f \rightarrow 0$. $\square$ as $z \rightarrow z_0$.

2.2.1 Thm. f holo in $D \subset \mathbb{C}$. $\Rightarrow \exists F$ primitive in D .

Pf. (i) $F(z) = \int_{\gamma_z} f(w) dw$ $w/ \nearrow$ path γ_z . by 2.1.1.
 (ii) for $z+h$ in D , $F(z+h) - F(z) = \int_{\gamma} f(w) dw$ for γ straight line.
 (iii) f cts $\Rightarrow f(w) = f(z) + \psi(w)$ ($\psi(w) \rightarrow 0$ as $w \rightarrow z$).
 (iv). $|\int_{\gamma} \psi(w) dw| \leq \sup |\psi(w)| |h| \rightarrow 0$ as $h \rightarrow 0$ so $F(z+h) - F(z) = \int_{\gamma} f(z) dw = f(z)h$ as $h \rightarrow 0$.
 (v). $F(z+h) - F(z) = \int_{\gamma} f(z) dw + \int_{\gamma} \psi(w) dw = f(z)h$ as $h \rightarrow 0$.

2.2.2 Thm. f holo on D . γ closed cur in D . $\int_{\gamma} f = 0$.

(Cauchy's).

2.4.1 Thm. f holo on $\mathbb{C} \supset \Omega \supset D$. $C = \partial D$. $\Rightarrow f(z) = \frac{1}{2\pi i} \int_C \frac{f(w)}{w-z} dw$ ($z \in D$).

Pf. $F(w) = f(w)/(w-z)$ holo on $\Omega - \{z\}$. $\Rightarrow \int_{\Gamma_\epsilon} F = 0$
 break up F as $\frac{f(w)-f(z)}{w-z} + \frac{f(z)}{w-z} \Rightarrow \int_{C_\epsilon} \frac{f(z)}{w-z} = -2\pi i f(z)$
 $\Rightarrow \int_{C_\epsilon} F = -\int_C F \Rightarrow 2\pi i f(z) = \int_C \frac{f(w)}{w-z} dw$ $\square$.

2.4.2 Cor. f holo in $\Omega \subset \mathbb{C}$. $\Rightarrow f$ has a complex deriv in Ω . and if $\bar{D} \subset \Omega$ then
 $f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\partial D} \frac{f(w)}{(w-z)^{n+1}} dw$

2.3.6 (Schwarz Reflector). f holo in Ω^+ , ext to I cts, f real on I
 $\Rightarrow F(z) = \overline{f(\bar{z})}$ h.o. on Ω^- . $\forall z_0 \in \Omega^-$, $f(\bar{z}) = \sum_{n=0}^{\infty} (\bar{z} - \bar{z}_0)^n$
 $\Rightarrow F(z) = \sum_{n=0}^{\infty} \overline{a_n} (z - z_0)^n \Rightarrow F$ h.o. $\Rightarrow$ symmetry principle so $F \circ f$ h.o.

2.4.3. Thm. (Cauchy's Inequality). f holo on $\Omega \supset \overline{D}_{z_0, R} \Rightarrow |f^{(n)}(z_0)| \leq \frac{n! \|f\|_{\partial D}}{R^n}$.

Pf. $|f^{(n)}(z_0)| = \left| \frac{n!}{2\pi i} \int_{\partial D} \frac{f(w)}{(w-z_0)^{n+1}} dw \right| \leq \frac{n!}{2\pi} \cdot \frac{2\pi R \cdot \|f\|_{\partial D}}{R^{n+1}} = \frac{n! \|f\|_{\partial D}}{R^n}$.

2.4.4. Thm. f holo on $\Omega \supset \overline{D}_{z_0}$ $\Rightarrow f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$ on D for $a_n = \frac{f^{(n)}(z_0)}{n!}$

Pf. Fix $z \in D$. $f(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w-z} dw$. $\frac{1}{w-z} = \frac{1}{w-z_0} \frac{1}{1 - \frac{z-z_0}{w-z_0}} = \frac{1}{w-z_0} \sum_{n=0}^{\infty} \left(\frac{z-z_0}{w-z_0} \right)^n$
 $|z-z_0| < R$ for $0 < n < \infty \Rightarrow$ geometric series $\sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{(w-z_0)^{n+1}} dw \right) \cdot (z-z_0)^n$.

2.4.5. Thm. (Liouville's). f entire and bounded $\Rightarrow f$ constant.

Pf. $|f'(z)| \leq \frac{B}{R}$ by Cauchy + Bound ($R \rightarrow \infty \Rightarrow f' = 0$ for all z since f entire) $\Rightarrow f$ const.

2.4.6. Cor. $P(z) = a_n z^n + \dots$ no non constant has a root in $\mathbb{C}$

~~Pf. $a_n \neq 0$. P no roots $\Rightarrow 1/P$ entire and bounded $\Rightarrow$ const $\Rightarrow P$ const.~~

2.4.8. Thm. f holo in Ω . f vanishes on dist $\{z_k\} \rightarrow z_0 \in \Omega \Rightarrow f \equiv 0$.

Pf. D cent @ z_0 inside Ω . power series $\sum a_n (z-z_0)^n$. if $f \neq 0$ on D , $\exists a_n \neq 0$ so $f(z) = a_n (z-z_0)^n (1+g(z-z_0))$. $g(z-z_0) \rightarrow 0$ as $z \rightarrow z_0$ so by cts $g(z_0) = 0$ but $1+g(z_0) \neq 0$ due $f(z_0) \neq 0$ so $f \neq 0$ on D .

Let $U = \{z: f(z) = 0\}^0$. U open and nonempty by cts. closed by cts. $U = \Omega$.

2.4.9. Cor. f, g holo in Ω and $f(z) = g(z) \forall z$ in nonempty open subset of $\Omega \Rightarrow f \equiv g$ on Ω .

Remk. $\text{res}_{z_0} f = \lim_{z \rightarrow z_0} (z-z_0) f(z)$. $\text{res}_{\infty} f = -\text{res}_0 \frac{1}{z} \cdot f(1/z)$

3.3.2 Remk. $\begin{matrix} \text{nonremovable} & \leftrightarrow & \text{bounded near } z_0 \\ \text{holo pole} & \leftrightarrow & |f(z)| \rightarrow \infty \text{ as } z \rightarrow z_0 \\ \text{essential} & \leftrightarrow & \text{else.} \end{matrix}$

$\uparrow$
poles @ 0 $\leftrightarrow$ poles @ ∞
of order 0

3.3.1  Then (Riemann's Removable Singularities). f holo on $\Omega - \{z_0\} \Rightarrow z_0$ removable.
 Pf. ~~f is bounded on $D \cap \Omega$. For $z \in D$, $f(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w-z} dw$.~~

$D \ni z_0$. $\bar{D} \subset \Omega$. Let $g(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w-z} dw$. $g \equiv f$ on $D - \{z_0\}$.
 $g'(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w-z} dw = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w-z_0} dw$ since g holo $\Rightarrow g$ entire.

3.3.3 Then (Casorati-Weierstrass). f holo on $D - \{z_0\}$ w/ essential sing at $z_0 \Rightarrow$ dense in $\mathbb{C}$.
 Pf. say not dense. $\exists w \in \mathbb{C}, \delta > 0$ so $|f(z) - w| > \delta$ on $D - \{z_0\}$.
 $\Rightarrow g(z) = \frac{1}{f(z) - w}$ holo on $D - \{z_0\}$ and bounded by $1/\delta \Rightarrow$ removable sing at z_0 .

rephrased. if $g(z_0) = 0$, ~~then~~ f has pole at z_0 , if $g(z_0) \neq 0$, f holo at $z_0 \Rightarrow$ $\checkmark$.

3.4.1 Then (Argument Principle) f meromorphic in $\Omega \subset \mathbb{C}$. if f 'nice' on ∂D , then
 $\frac{1}{2\pi i} \int_{\partial D} \frac{f'}{f} = (\# \text{ zeros inside } D) - (\# \text{ poles inside } D)$ counting multiplicities.
 Pf. note $\left(\frac{f_1 f_2}{f_1 f_2}\right)' = \frac{f_1'}{f_1} + \frac{f_2'}{f_2}$ remember $\frac{f'}{f} = \frac{(\pi f')}{\pi f}$. if $f(z) = (z - z_0)^n g(z)$ holo, $\frac{f'}{f} = \frac{n}{z - z_0} + \frac{g'}{g}$ (or $-n$ if pole).

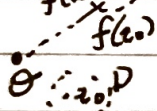
3.4.3 Then (Rouché) f, g holo in $\Omega \subset \mathbb{C}$. if $|f| > |g|$ on $\partial D \Rightarrow f, f+g$ have same # of zeros inside D .
 Pf. for $t \in [0, 1]$, $f_t \equiv f + tg$. let $\eta_t =$ zero of f_t w/ multiplicity inside D .
 $|f| > |g| \Rightarrow f_t$ no zeros on ∂D . ~~argument principle~~ $\eta_t = \frac{1}{2\pi i} \int_{\partial D} \frac{f'_t}{f_t}$ as t in t $\Rightarrow$ η_t is constant for f_1, f_0 .

3.4.4 Then (Open) f holo and non-const $\Rightarrow f$ open.

Pf. ~~say~~ say $w_0 = f(z_0)$. let $g = f - w_0 = (f - w_0) + (w_0 - w_0)$ $\forall w$ near w_0 .
 $\Rightarrow$ pick $\delta > 0$ so $|z - z_0| \leq \delta \subset \Omega$ and $f(z) \neq w_0$ on $|z - z_0| = \delta$.

by non-const. pick $\epsilon > 0$ so $|f(z) - w| \geq \epsilon$ on $|z - z_0| = \delta$ if $|w - w_0| < \epsilon$, $|f - w_0| > |w_0 - w|$.
 $\Rightarrow$ Rouché so g has a zero inside $B_\delta(z_0)$ $\forall w$ near w_0 . $\square$

3.4.5 Then (Maximum Modulus Principle). f non-const holo in $\Omega \Rightarrow f$ cannot attain max in Ω .
 Pf. say f attained max at z_0 . f holo $\Rightarrow f$ open $\Rightarrow B_\epsilon(z_0) \subset \Omega \Rightarrow f(B_\epsilon(z_0))$ open and contains $f(z_0)$. ~~clay~~ $\exists z \in D$ so $|f(z)| > |f(z_0)|$. $\square$



3.4.6 Cor. Ω region w/ cpt $\bar{\Omega}$. f holo on Ω , cts on $\bar{\Omega} \Rightarrow \sup_{\bar{\Omega}} |f| \leq \sup_{\partial\Omega} |f|$.
 Pf. f const: trivial. f non-const: f cts on $\bar{\Omega} \Rightarrow$ attains max on $\bar{\Omega}$. f holo on $\Omega \Rightarrow f$ no max on Ω . f p.h. $\nrightarrow \infty$.

NOTES Rnk. $f(z) = \lambda \prod_{j=1}^N (z - z_j)^{m_j} \Rightarrow F(z) = f(1/z) = \lambda \frac{\prod_{j=1}^N (1 - z z_j)}{z^N}$ pole of NE 0

NOTES Prop. $f: \mathbb{C} \rightarrow \mathbb{C}$ holo, $\exists N, C$ s.t. $|f(z)| \leq C|z|^N \forall z$ in punctured nbhd of ∞ .
 $\Rightarrow f$ is a polynomial of deg $\leq$ most N .
 Pf. wlog $f^{(N+1)} \equiv 0$ for $n = N+1$, r large enough s.t. Laurent applies.
 $\Rightarrow |f^{(N+1)}(z)| \leq \frac{(N+1)!}{r^{N+1}} \sup_{|z|=r} |f| \leq \frac{(N+1)!}{r^{N+1}} \cdot C \cdot (|z| + r) \xrightarrow{r \rightarrow \infty} 0$. $\square$

NOTES Cor. $f: \mathbb{C} \rightarrow \mathbb{C}$ holo w/ pole at $\infty \Rightarrow f$ polynomial.
 Pf. wlog $\exists N, C$ s.t. $|f(z)| \leq C|z|^N$ on nbhd of ∞ .
 Let $F(z) = f(1/z)$. F p.h. at $0 \Rightarrow F(z) = \frac{a_{-N}}{z^N} + \dots + a_0$ about 0.
 $\Rightarrow |f(z)| = |F(1/z)| \leq C|1/z|^N$ for z close to 0. (a-m to 0).
 $\Rightarrow |f| \leq C|z|^N$ for $|z|$ large enough. $\Rightarrow$ last thm $\square$.

NOTES Defn. A rational function is ratio of polynomials $f(z) = P(z)/Q(z)$.

NOTES Thm. $f: \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ meromorphic $\Rightarrow f$ rational.
 Pf. $\hat{\mathbb{C}}$ cpt $\Rightarrow$ singularities isolated & finite, say $z_1, \dots, z_N$, and maybe $z = \infty$.
 $z \neq z_j \Rightarrow f$ meromorphic at $z_j \Rightarrow \exists n_j$ s.t. $(z - z_j)^{n_j} f(z)$ extends holo to z_j .
 $\Rightarrow f(z) = \prod_{j=1}^N (z - z_j)^{-n_j}$ holo h.t. ∞ and meromorphic at ∞ . $\Rightarrow$ poly $\frac{f(z)}{\prod_{j=1}^N (z - z_j)^{n_j}}$ is const.

Def. Bijective holo $f: U \rightarrow V$ is conformal map. U and V are conformally equivalent or biholomorphic.

Rmk. f bi-holo $\Rightarrow f^{-1}$ holo.

8.1.2 $\overset{\text{Thm. 6}}{\underset{\text{of } \mathbb{C}}{H}} \xrightarrow[F]{G} D$, $F = \frac{i-z}{i+z}$, $G = i \frac{1-w}{1+w}$ conformal mapping

Pf. (i) F, G both h/b in own domains, (ii) $|F| < 1$,
 (iii) $\text{Im}(G) = \text{Re}\left(\frac{1-u-i v}{1+u+i v}\right) = \frac{1-u^2-v^2}{(1+u)^2+v^2} > 0$ ($u^2+v^2 < 1$).
 (iv). $F \circ G = \text{id}_D$ $G \circ F = \text{id}_H$.

1 RMK. F maps $\mathbb{R}$ into ∂D . $z \mapsto \frac{az+b}{cz+d}$ fractional linear transform

NOTES $GL_2(\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{C}, ad-bc \neq 0 \right\}$.

$SL_2(\mathbb{C}) = \det^{-1}(1)$, $GL_2(\mathbb{C}) \xrightarrow{\det} \mathbb{C}^*$

$GL_2(\mathbb{C}) \curvearrowright \hat{\mathbb{C}} : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az+b}{cz+d}$ Möbius transform.

$\begin{pmatrix} \lambda & 0 \\ 0 & 1 \end{pmatrix} \cdot z = \text{id} \cdot z \Rightarrow GL_2(\mathbb{C}) = PGL_2(\mathbb{C})$

$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \cdot z = z+b$ $SL_2(\mathbb{C})/\pm \text{id} = PSL_2(\mathbb{C})$

$\begin{pmatrix} \lambda & 0 \\ 0 & 1 \end{pmatrix} \cdot z = \lambda z$

Riemann sphere: $\hat{\mathbb{C}} \simeq \mathbb{C}^2 - \{0,0\} / \mathbb{C}^* = \{(z, \omega) : z \text{ and } \omega \text{ not both } 0\}$
 $\omega \neq 0 \Rightarrow$ act by $1/\omega \Rightarrow (z/\omega, 1) \Rightarrow$ copy of $\mathbb{C}$.
 $\omega = 0 \Rightarrow$ act by $1/z \Rightarrow (1, 0) \Rightarrow \infty \rightarrow \hat{\mathbb{C}}$

$GL_2(\mathbb{C}) \curvearrowright \mathbb{C}^2 - \{0,0\} : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} z \\ \omega \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} z \\ \omega \end{pmatrix}$

$\dim_{\mathbb{C}} = 3$

PROP. $PSL_2(\mathbb{C})$ acts triply transitively on $\hat{\mathbb{C}}$, i.e. u, v, w distinct u', v', w' distinct g_0, g_1, g_2

Pf. we will show for $u'=0, v'=1, w'=\infty$ exist g .

(i) $\omega \rightarrow \infty$ by $\frac{1}{\omega}$ $\omega \neq \infty$ $\omega = \infty$ (ii) $u \rightarrow 0$ by $z \mapsto z-u$

unique, any $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ from $(0,1,\infty) \rightarrow \infty = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \infty = \frac{a}{c} \Rightarrow c=0$

$\Rightarrow 0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot 0 = \frac{b}{d} \Rightarrow b=0$

$\Rightarrow 1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot 1 = \frac{a+d}{d} \Rightarrow a=d$ a.

$u \quad v \quad w \quad \frac{1}{\omega}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $u_1 \quad v_1 \quad \infty \quad \frac{1}{\omega}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $0 \quad u_1 \quad \infty \quad \frac{1}{\omega}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $0 \quad 1 \quad \infty \quad \frac{z}{v_2}$

NOTES

LEMMA. $g \neq id \in PSL_2(\mathbb{C}) \Rightarrow g$ 2 fixed pts of $\mathbb{C}$ (w/ multiplicity).

Pf. $\hat{G} = e^{i\theta} \begin{pmatrix} a & b \\ c & d \end{pmatrix} / \det$, then that fixed pts $\Leftrightarrow c$ -value (invertible 2×2 matrix...)

parabolic
(translation)

EX. $z \mapsto z + t$ for $t \in \mathbb{C}$ if w/ multiply $z \mapsto \frac{z+t}{1}$ 

(rotation)
elliptic

EX. $\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \in PSL_2(\mathbb{C})$ $z \mapsto e^{i\theta} z$, fixed pts 0 and ∞ 

(dilation)
hyperbolic

EX. $\begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} \in PSL_2(\mathbb{C})$ $z \mapsto e^t z$, fixed pts 0 and ∞ 

RMK. in $\mathbb{C}$, $\exists$ unique line thgh 2 pts, circle thgh 3 pts

DEF. $U \subset \mathbb{C}$, $Aut(U) = \{ \text{biholom } U \rightarrow U \}$.

PROP. $Aut(\mathbb{C}) \cong PSL_2(\mathbb{C})$

Pf. $f \in Aut(\hat{\mathbb{C}})$. $z_0 = f(\infty)$. pick $g \in PSL_2(\mathbb{C})$ mapping z_0 to ∞ .

$\Rightarrow g \circ f \in Aut(\hat{\mathbb{C}})$ maps $\infty \rightarrow \infty$. so polynomial so bijection $\hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ after restriction

so linear $\Rightarrow h = g \circ f$ Möbius $\Rightarrow f = g^{-1} \circ h$ Möbius.

$H = \{ z \in \mathbb{C} : \text{Im } z > 0 \}$.

LEMMA. if $f \in Aut(H)$ extends to $Aut(\hat{H})$, then f extends to $Aut(\hat{\mathbb{C}})$.

punctured
disk D^*

Pf. Schwarz $\Rightarrow f$ extends to $Aut(\mathbb{C})$. WTS f maps ∞ to ∞ .

$U \cap V = \emptyset$

has $f(z)$ given by linear poly. if not, then f has essential sing at ∞ so dense (Casorati-Weierstrass) and $f(U)$ is open int $\mathbb{C}$ since f inj. so f maps, not essent.

LEMMA (Schwarz). $f: D_r(0) \rightarrow D_1(0)$ holom on $f(0) = 0$.

then (i) $|f(z)| \leq |z|$ for all $z \in D_r(0)$. (ii) $|f(z)| = |z|$ for some $z_0 \in D_r(0) \setminus \{0\} \Rightarrow f$ is a rotation
(iii) $|f'(0)| \leq 1$ ~~if $|f'(0)| = 1$ then f is a rotation~~

Pf. f zero at 0 of order $\geq 1 \Rightarrow f(z)/z$ removable sing at 0. $h(z) = \text{holo. extn. of } f/z$.

(i) $\forall r < 1, |h(z)| = \frac{|f(z)|}{|z|} \leq \frac{1}{r} \forall z \in \partial_r(0) \Rightarrow |h(z)| \leq \frac{1}{r} \forall z \in D_r(0)$
 $\lim_{r \rightarrow 1} \frac{1}{r} = 1$ gives $|h(z)| \leq 1 \Rightarrow |f| \leq |z|$.

(ii) $|f(z)| = |z| \Rightarrow |h(z)| = 1 \Rightarrow h(z) = e^{i\theta} \Rightarrow f(z) = e^{i\theta} z$.

(iii) $f'(0) = \lim_{z \rightarrow 0} \frac{f(z)}{z} = h(0) \Rightarrow |f'(0)| = |h(0)| = 1 \Rightarrow f(z) = e^{i\theta} z$.

we're done $H \sim ID$. vts $Aut(H) \sim Aut(ID)$.

NOTES

Lemma. $PSL_2(\mathbb{R})$ acts transitively on H . i.e. $\exists g \in PSL_2(\mathbb{R})$ s.t. $g \cdot v_1 = v_2$
 Pf. $v_1 = x + iy$, $0 < y = 1$. let $b = -x$ so $z \mapsto z + b$ takes v_1 to iy .
 let $a = 1/y$ so $z \mapsto az$ takes iy to i , $a > 0$ so $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \in PSL_2(\mathbb{R})$. $\square$

Prop. $g \in Aut(H)$ acts $\dagger$ $Aut(\hat{\mathbb{C}})$ so given by moebius.

Pf. since $D \sim H$, it's show $Aut(D)$ acts $\dagger$ $Aut(\hat{\mathbb{C}})$.
 (by moebius!)

$g \in Aut(D)$, $z_0 = g(0)$. $PSL_2(\mathbb{R}) \subset PSL_2(\mathbb{C})$ acts transitively on H .

so $\exists$ moebius transfer $h \in Aut(D)$ so $h: z_0 \mapsto 0$. $h \circ g \in Aut(D)$ sends $0 \mapsto 0$

Schwarz lemma $\Rightarrow |G| \leq |z|$ and if equality then h is a rotation

let $h \circ g$ biholom on D so is $g^{-1} \circ h^{-1} \Rightarrow |G| \leq |z|$
 so $|z| = |G(z)| \Rightarrow h$ rotation. $\Rightarrow$ extends $\square$

DEF. $\mathcal{F}$ family of fcts on U normal if every seq $\{f_n\}$
 has a uniformly convg subseq on all gt sets of U .

DEF. K cpt. $\mathcal{F}$ uniformly bounded on K if $\exists B \in \mathbb{R}$ so $|f| < B \forall z \in K, \forall f \in \mathcal{F}$.

DEF. $\mathcal{F}$ equicontinuous if $\forall \epsilon > 0, \exists \delta > 0$ so $|f(z_1) - f(z_2)| < \epsilon$
 $\forall |z_1 - z_2| < \delta, \forall f \in \mathcal{F}$.

Thm (Arzela - Ascoli). $\mathcal{F}$ equicontinuous and uniformly bounded on cpt K
 Pf. (i) let $U = \bigcup_{n=1}^{\infty} K_n$ where $K_n = \text{Int}(K_{n+1})$. $\Rightarrow$ normal.

(ii) on any $K = K_n$, pick countable dense seq. in K $\{z_n\}$.
 take seq $\{f_{n_j}\} \in \mathcal{F}$. $\{f_{n_j}(z_1)\}_m$ bounded $\Rightarrow \{f_{n_j}(z_1)\}_m$ conv.
 apply for each z_n . $\Rightarrow \exists$ subseq $f_{n_j}, \dots$ conv. @ z_n .
 doing

(iii) wts f_{n_j} conv. unifly on K . $\forall \epsilon > 0, \{z_n\}$ dense in K $\forall z \in K$,
 $\exists z_n$ s.t. $|z - z_n| < \delta \Rightarrow |f_{n_j}(z) - f_{n_j}(z_n)| \leq |f_{n_j}(z) - f_{n_j}(z_n)|$

$2nd: \{f_{n_j}\} \subset \mathcal{F}$ conv. unifly on K $\Rightarrow \exists f$ s.t. $|f_{n_j}(z) - f(z)| < \epsilon/3$
 $\{f_{n_j}\} \subset \mathcal{F}$ conv. unifly on K $\Rightarrow \exists f$ s.t. $|f_{n_j}(z) - f(z)| < \epsilon/3$
 so normal. $\square$

$Aut(H) \sim PSL_2(\mathbb{R})$.

Pf. $f \in Aut(H)$.

$\psi: D \rightarrow H, g: H \rightarrow D, \phi: D \rightarrow ID$

Since g rotation/mobius $\Rightarrow$ fctive.

$0 \mapsto 0$

$g = \text{pop of } 0 \in Aut(D)$ sends $0 \mapsto 0$

$g(f(v(0))) \rightarrow 0$

NOTES. Thm. (Montel's). $\mathcal{F}$ family of holom on U , unifly bound on opt sub. of U .

$\Rightarrow$ $\mathcal{F}$ equicontinuous (by Arzela-Ascoli) normal.

Pf. suffices to show $\{f'(z) : f \in \mathcal{F}\}$ unifly bounded opt sub.

pick $K \subset U$ opt. dist $(K, \mathbb{C} \setminus U) > 0$ so $\exists k_g > k$

s.t. $B_g(z) \subset K_g \forall z \in K$. $|f(z_1) - f(z_2)| = |\int_{z_1}^{z_2} f'(z) dz|$

$\Rightarrow$ ok to bound $|f'(z)|$ unifly on K .

(Cauchy
inty)

$\Rightarrow |f'(z)| \leq \frac{1}{2\pi g} \sup_{\partial B_g(z)} |f(z)|$. $\partial B_g(z) \subset K_g$ so bounded unifly on K .

Prop. $U \subset \mathbb{C}$ connect and $\{f_n : U \rightarrow \mathbb{C}\}_{n=1}^\infty$ seq. of inj. holom. conv. unifly to f on opt sub.

Pf.

Idea: $\int_{f_n - w}^{f'_n} \rightarrow \int_{f - w}^{f'}$ counts pre-images.

the f is inj. on opt sub.

say f not inj, so $\exists z_1 \neq z_2$ s.t. $f(z_1) = f(z_2)$. take $g_n(z) = f_n(z) - f_n(z_2)$.

then, $g(z) := \lim_{n \rightarrow \infty} g_n = f(z) - f(z_2)$ zero at z_1 .

(unifly on opt sub)

g non const $\Rightarrow$ zero isolated $|g| \neq 0$ on $B_\epsilon(z_1) \setminus \{z_1\}$ for some ϵ .

$\Rightarrow \frac{1}{2\pi i} \int_{\partial B_\epsilon(z_1)} \frac{g'}{g} \in \mathbb{Z} \neq 0 = \lim_{n \rightarrow \infty} \int_{\partial B_\epsilon(z_1)} \frac{g'_n}{g_n} \Rightarrow f_n(z) \neq f_n(z_2)$ zero at z_1

$\square$. g and f const. $\Leftarrow f$ not inj $\Leftarrow$ f not inj. $\Leftarrow$ f not inj.

Thm. (Riemann Mapping). $U \subset \mathbb{C}$ simply connect. $z_0 \in U$. $\exists$ unique biholom $f: U \rightarrow \mathbb{D}$ s.t. $f(z_0) = 0$, $f'(z_0) > 0$.

Pf. STEP 1 find bihol. $U \rightarrow \mathbb{D}$

Translate so $0 \notin U$. Scale so $z_0 \in U$.

Using conn $\Rightarrow \exists \log_U$. $e^{i \log_U(z)} = z \Rightarrow \log_U$ inj. $\Rightarrow \log_U(z) \neq 2\pi i \forall z \in U$.

Also, $|\log_U(z) - 2\pi i|$ bounded below since $\log_U(z_0)$ since $e^{2\pi i} = 1$ bounded inj.

$\Rightarrow F(z) = \frac{1}{\log_U(z) - 2\pi i}$ bounded and inj.

$\Rightarrow$ scale and translate so contin. $\mathbb{D}$ al sends $z_0 \mapsto 0 \in \mathbb{D}$.

STEP 2 $\mathcal{F} = \{f: \mathbb{D} \rightarrow \mathbb{D} : f(0) = 0, f \text{ holom, inj}\}$. $\mathcal{F}$ unifly bound so $\{f'(0)\}$ bound let $C = \sup |f'(0)|$, take seq f_n so $|f'_n(0)| \rightarrow C$.

Montel's Thm $\Rightarrow \exists$ conv. subseq w/ holom limit f .

f sur

STEP 3 f sur. $\forall \alpha \in \mathbb{D}, \psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}$ auto of $\mathbb{D}$ $0 \leftrightarrow \alpha$ s.t. $\psi_\alpha = \psi_\alpha^{-1}$

$(\psi_\alpha(e^{i\theta})) = 1 \Rightarrow$ minimal $\psi_\alpha: \mathbb{D} \rightarrow \mathbb{D}$. $F = \psi_\alpha \circ f \circ \psi_\alpha^{-1}$ sends 0 to α and is inj.